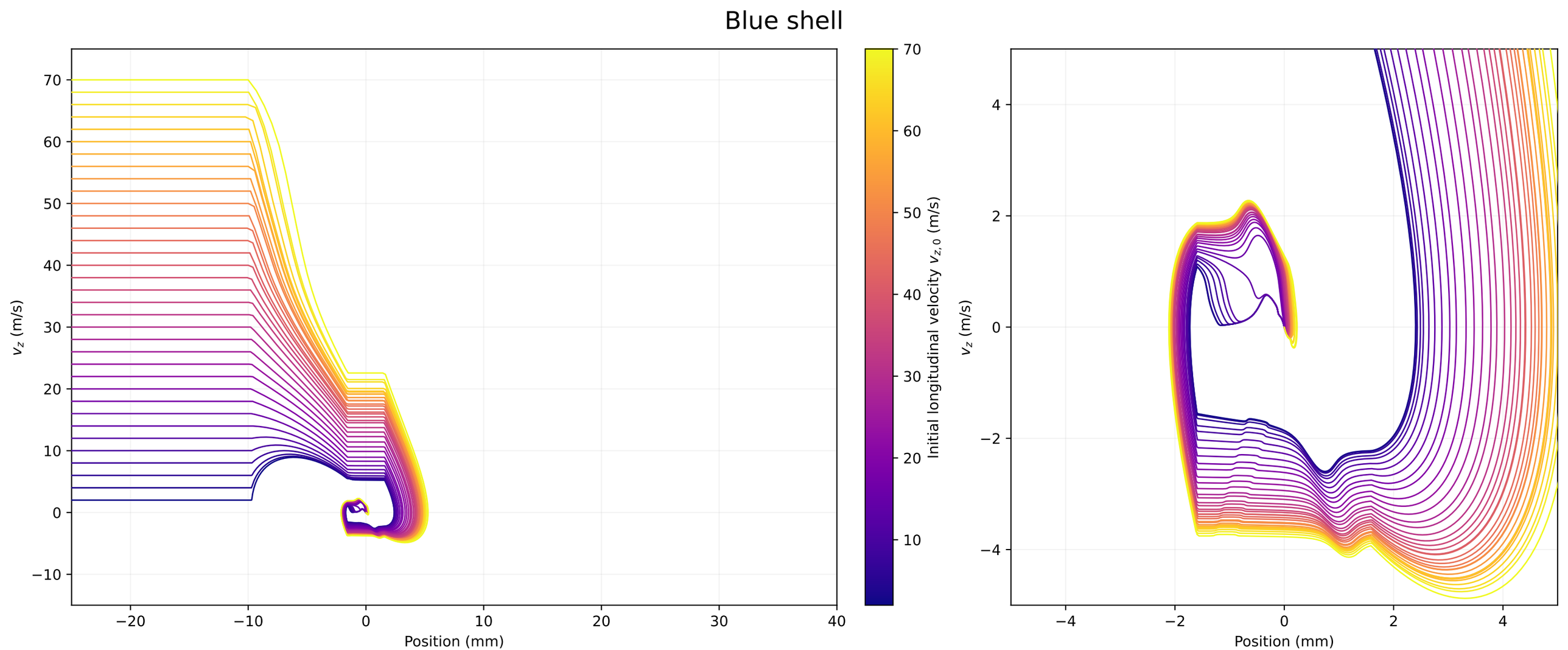

Blue shell
$v_z$ (m/s)
Position (mm)
Initial longitudinal velocity $v_{z,0}$ (m/s)
Central longitudinal slice: $x = y = v_x = v_y = 0$ · deterministic mean-force trajectories
End-of-run classification at 100 ms: distance ≤ 5 mm and speed ≤ 1 m/s (1D model).
Both panels show the same trajectories; the right panel is a central zoom.